IERG4320/IEMS5726: Data Science in Practice

Lecture 1A: Course Introduction

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The Chinese University of Hong Kong

Outline

1. Course Administration
2. Introduction to Data Science
3. Reference Books
4. Python Programming Environment

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The Department of Information Engineering, CUHK

People

Lecturer Danny Yip @ SHB735 Office Hour: On appointment

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For lecture content, materials, clarifications of course project, reference materials, …

Tutor 1 YANG Bufang @ SHB702

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Contact Bufang if you need specific help when working on your assignment 2 and 3

Tutor 2 WANG, Genglin @ SHB 828

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Contact Genglin if you need specific help when working on your assignment 1 and 4

Lectures

- Main Lecture
  - Wednesday 7:00 - 10:00 pm @ LSB LT6
- Dates (12 Lectures)
  - 7/1, 14/1, 21/1, 28/1, 4/2, 11/2, 25/2, 4/3, 11/3, 18/3, 25/3, 1/4
- No tutorial components in master course
- Final Examination will be week 15
  - Location: ERB 1004 (IE lab)
  - Date: 15 Apr (Wed)
  - Time: Session A1: 6:00 - 8:00 pm, Session A2: 8:30 - 10:30 pm
  - Open book, open note, close internet, close discussion
- Notes
  - No classes on 18 Feb due to Chinese New Year Holiday

Course Materials

- All course materials are uploaded onto the Blackboard
- http://blackboard.cuhk.edu.hk/
  - General information
  - Announcement
  - Lecture notes and recordings
  - Assignments and course project
- Suggested solutions will not be available
- Please submit all your coursework to the Blackboard. We do not accept coursework via email attachments.

Course Schedule - Lecture

#	Topic	Date
1	Course Outline, FastAPI App	7/1
2	Basic Statistics	14/1
3-5	Data Representation	14, 21, 28/1
6-7	Data Modeling	4, 11/2
8	Data Visualization	25/2

Course Schedule - Lecture

#	Topic	Date
9	Case Study I: Business	4, 11/3
10	Case Study II: Text mining	11, 18/3
11	Case Study III: Multimedia	18, 25/3
12	Case Study IV: Game Applications	25/3, 1/4
13	Course Summary	1/4
#	Final Examination	15/4

Course Assessment

1. **UReply attendance (5%)**
   - Ureply question(s) available during second break (MSc session)
   - Participate one UReply activity = 0.5%
   - No matter your answer is correct or not
   - Remember to backup the questions on your own
2. **4 Assignments (35%)**
   - Both written and programming practices
   - We use Python programming language for the programming exercises
   - Deadlines: 22 Jan (Thur), 5 Feb (Thur), 26 Feb (Thur), 12 Mar (Thur)
3. **Course project (20%)** due on 8 May (Fri)
4. **Final examination (40%)**
   - Written exam, open books/notes, close internet/discussion
   - Date: 15 Apr (Wed), Location: ERB 1004
   - Session A1: 6:00 - 8:00 pm, Session A2: 8:30 - 10:30 pm

More about the course project

- Decide your own data science course project
- Understand the steps in data science pipeline
- Deploy a service using both Python programming and deep learning

You will fail the course if …

1. Your final examination performance is unsatisfactory (e.g. below 30%)
2. Your grand total cannot meet the passing line (e.g. below 40%)
3. You do not submit the course project

No cheating, no plagiarisms

University Guidelines to Academic Honesty

- http://www.cuhk.edu.hk/policy/academichonesty/

Student/Faculty Expectations on Teaching and Learning

- https://www.cse.cuhk.edu.hk/~cslui/student_teacher_expectations.pdf

Use of ChatGPT, DeepSeek, Gemini (generative AI tools): Approach 3

- By default, generative AI tools are not allowed in all courses in the CUHK
- For this course, we release this constraints, you can use ChatGPT, Gemini, DeepSeek for the whole course (except the final exam)
- When you use the tools, write clearly which part, and how much you have used the tools in you work

Expectations from this course

1. Have basic understanding or **willing to learn** the Python **Programming** Language
2. Would like to **challenge** yourself **with** interesting programming and problems related to **different types of data**

Approach of this course

- Machine learning is a huge topic, to be good at machine learning, you will have to have good foundations in
  - Statistics
  - Linear algebra
  - Calculus
  - Probability theory
  - Signal processing
  - ...
- However, this is NOT just a course that follow the usual path of introduction to machine learning

Approach of this course

- In this course, we also focus on the following topics:
  - Data collection and preprocessing
  - Data representation
  - Data visualization
  - Problem specific domain
- What you will **learn** after taking this course?
  - Working on common machine learning tasks
  - Paying attention to the details to different problem domains
  - Visualizing the data at different stages
  - The intuition of neural representation of the data

Have your heard these terms before? What are they?

1. Data science
2. Data analysis
3. Big data
4. Database
5. Data mining
6. Machine learning
7. Deep learning
8. Artificial intelligence
9. Data scientist
10. Machine learning engineer
11. Data engineer
12. … etc

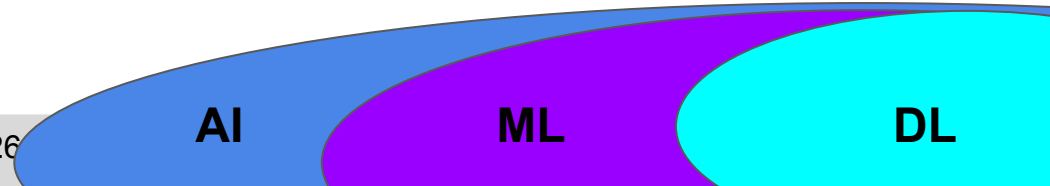
Terms related to “Data”

1. Database
   - A collection of data that is well-organized and easily managed
2. Data analysis
   - A process of retrieving knowledge from data
3. Data mining
   - A process of extracting knowledge in large data sets
4. Big data
   - A field that handle large and complex data sets
   - Challenges: data storage, data analysis, transferring data, searching data, visualizing data, updating data, … etc

Artificial Intelligence, Machine Learning, Deep Learning

- These are commonly used in data analysis

1. Artificial intelligence
   - Knowledge or behaviour demonstrated by machines
2. Machine learning
   - The study of computer algorithms that improve knowledge retrieval from data
   - A part of artificial intelligence
3. Deep learning
   - The use of multiple layers in the network to improve knowledge retrieval from data
   - A part of machine learning



Data Scientist, ML Engineer, Data Engineer

- These are roles, jobs

1. Data scientist transforms data to valuable insights and strategic decisions.
2. Machine learning engineer turns data-driven models into practical applications.
3. Data engineer works on the the foundation of a robust data infrastructure.

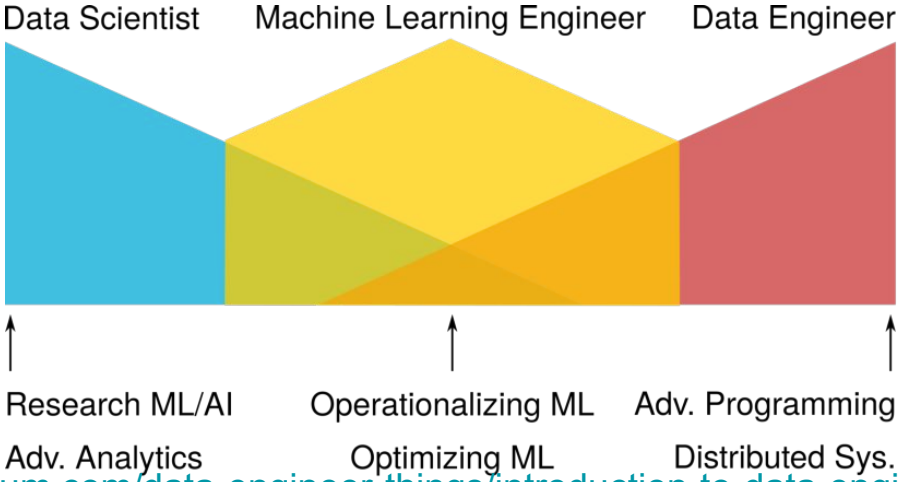


Image credit: https://medium.com/data-engineer-things/introduction-to-data-engineering-e16c9942dc2c

Data Science

- It uses scientific methods, process, algorithms and systems to extract knowledge and insights from data
- It applies knowledge and actionable insights from data across a broad range of application domains
- Relevant disciplines include:
  - Computer science - database, data processing, programming
  - Informatics - study of computation systems, knowledge extraction
  - Statistics - interpretation of the results
  - Other disciplines - depend on the problem definitions
  - Communication and visualization

Introduction to Data Science

Stages in Data Science Lifecycle

1. Data acquisition
2. Data cleaning and data warehousing
3. Data modeling and data summarization
4. Data analysis
5. Data visualization

Example 1: Handwritten Digit Recognition

- Recognize handwritten digits using a machine learning approach (one of the first widely used ML systems)

Reference: http://yann.lecun.com/exdb/mnist/
http://yann.lecun.com/exdb/publis/pdf/matan-92.pdf

Example 1: Handwritten Digit Recognition

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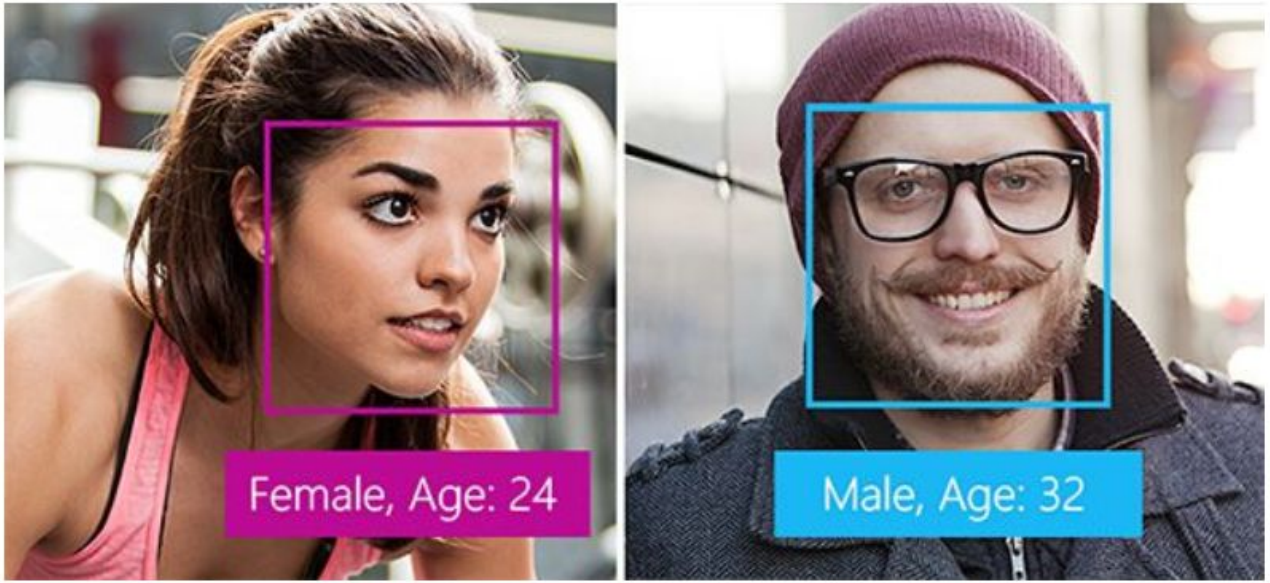
Reference: http://yann.lecun.com/exdb/mnist/
http://yann.lecun.com/exdb/publis/pdf/matan-92.pdf

Example 2: Recommender Systems

- Learn to recommend movies or books to users based on their past purchase and browsing behaviour
- The Netflix Prize: a competition to build a movie recommender system for Netflix
- The winning algorithm: BellKor's Pragmatic Chaos

Example 3: Face Detection / Recognition

- Learn to detect human faces in an image or a video stream
- Recognize the persons with the faces
- Example: Microsoft's Face API



Example 4: Machine Translation

- Statistical machine translation (SMT) or Neural machine translation (NMT) learn to translate sentences from one language into another language by analysing parallel text corpora
- Google's Neural Machine Translation: https://ai.google/research/pubs/pub45610

Example 5: Face Generator

- Generative Adversarial Network generates fake images through learning the features from the existing images.

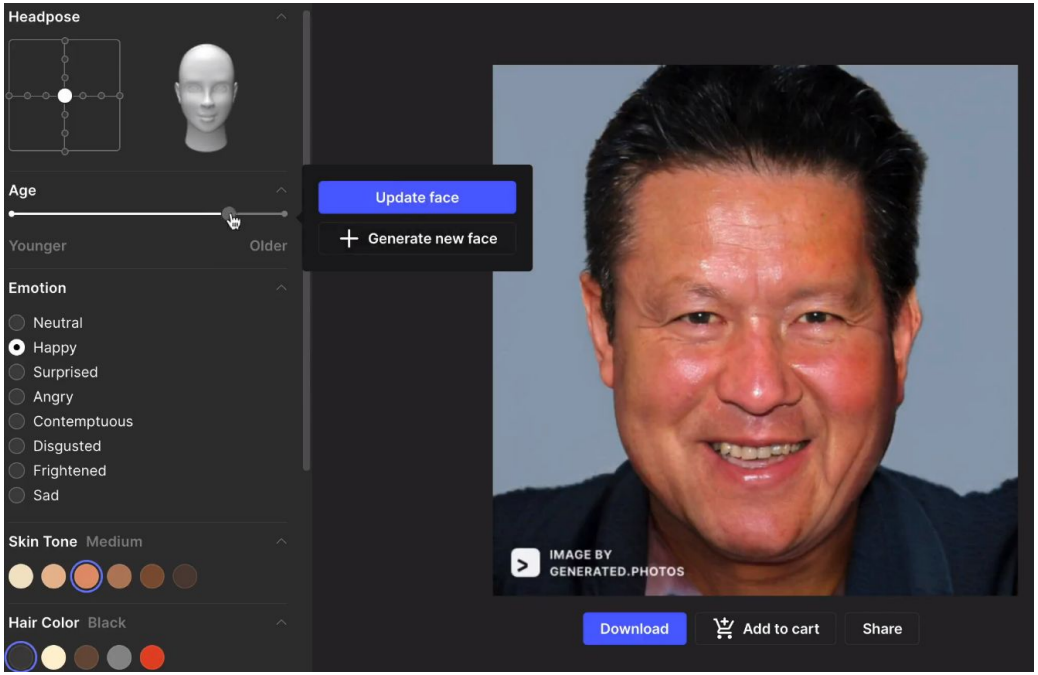


Image credit: https://generated.photos/face-generator

Example 6: ChatGPT

- ChatGPT is a natural language processing (NLP) tool driven by AI technology that allows you to have human-like conversations and much more with the chatbot. The language model can answer questions and assist you with tasks, such as composing emails, essays, and code.

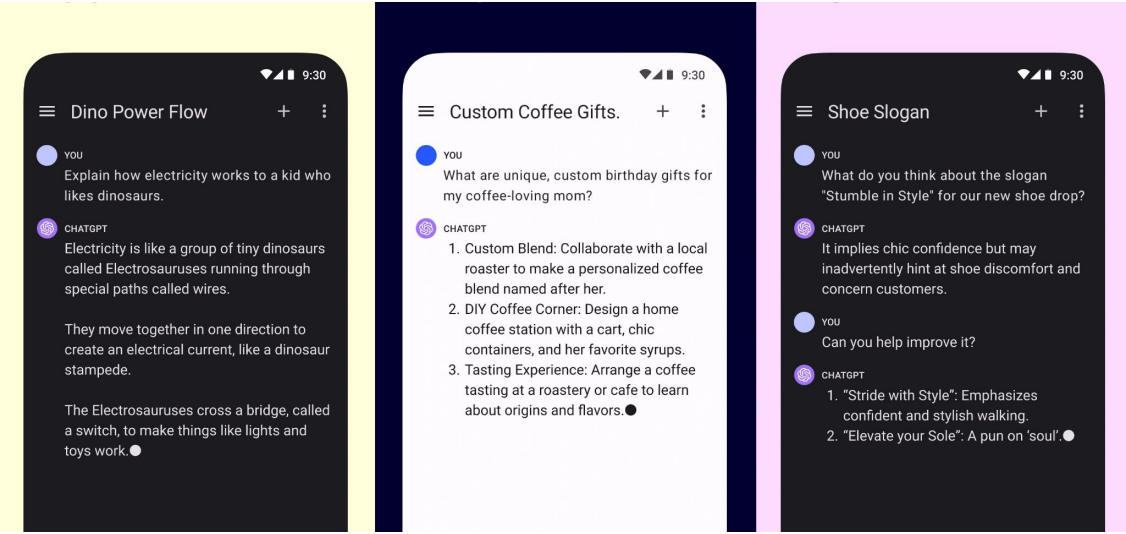


Image credit: TechCrunch

Main Types of Data Analytics

1. **Predictive analytics**: To make predictions about future trends or events and answers the question i.e. “What might happen in the future?”
2. **Descriptive analytics**: To pull trends from raw data and succinctly describe what happened or is currently happening, i.e. “What happened?”
3. **Prescriptive analytics**: To take into account all possible factors in a scenario and suggest actionable takeaways i.e. “What should we do next?”
4. (Optional) **Diagnostic analytics**: To compare coexisting trends or movement, uncovering correlations between variables, and determining causal relationships where possible, i.e. “Why did this happen?”

Example: Data Analytics - Ice-cream Shop

1. **Predictive analytics**: “What might happen in the future?”
   - How much chocolate ice-cream will be needed tomorrow?
2. **Descriptive analytics**: “What happened?”
   - What are the top favours in the past week?
3. **Prescriptive analytics**: “What should we do next?”
   - To make more money, should we increase or reduce the quality of the chocolate ice-cream?
4. (Optional) **Diagnostic analytics**: i.e. “Why did this happen?”
   - Why are chocolate and vanilla flavours the best sellers in the past week?

Importance of Data Science

1. Cost efficient
2. Serves as a decision making tool
3. Helps to understand customer behaviour
4. Analyze and manage risks
5. Product development and innovations

Are there any downside of the Data Science?

EU AI Act

- First Regulation on Artificial Intelligence
- Ensure that AI systems in the EU are safe and respect fundamental rights and values
  - Foster investment and innovation in AI
  - Enhance governance and enforcement
  - Encourage a single EU market for AI

Reference:
https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence

Different Levels of Risks

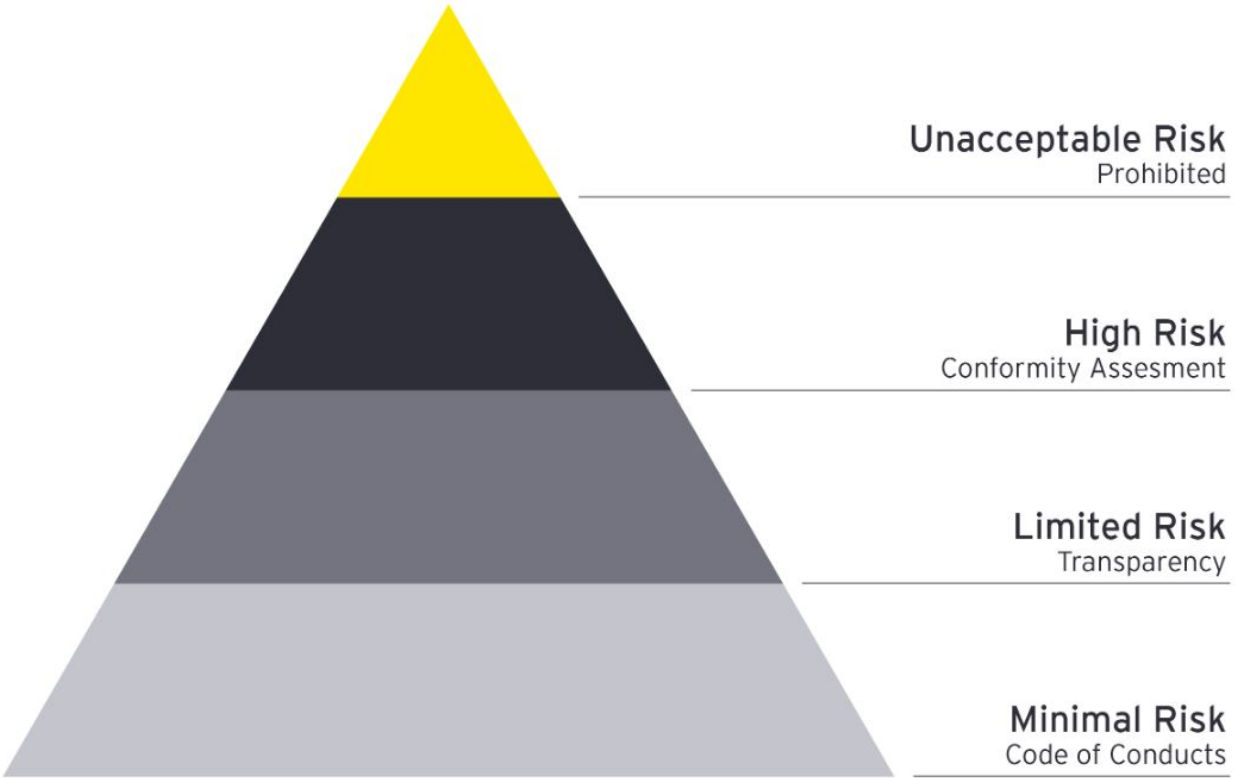


Image credit:
https://www.ey.com/en_ch/forensic-integrity-services/the-eu-ai-act-what-it-means-for-your-business

Different Levels of Risks

Unacceptable Risks: (Prohibited)

- Voice-activated toys that encourage dangerous behaviour in children
- Biometric identification
- Social scoring
- Deceptive techniques
- Facial recognition databases
- Exception: law enforcement purposes

High Risks: (Conformity Assessment)

- AI systems in cars, medical devices
- AI systems used in education, vocational training, employment, … etc
- AI system for record-keeping

Reference: https://artificialintelligenceact.eu/high-level-summary/

Different Levels of Risks

Limited Risks: (Transparency)

- ChatGPT
- Deep Fakes
- Chat bots

Minimal Risks: (Code of Conducts)

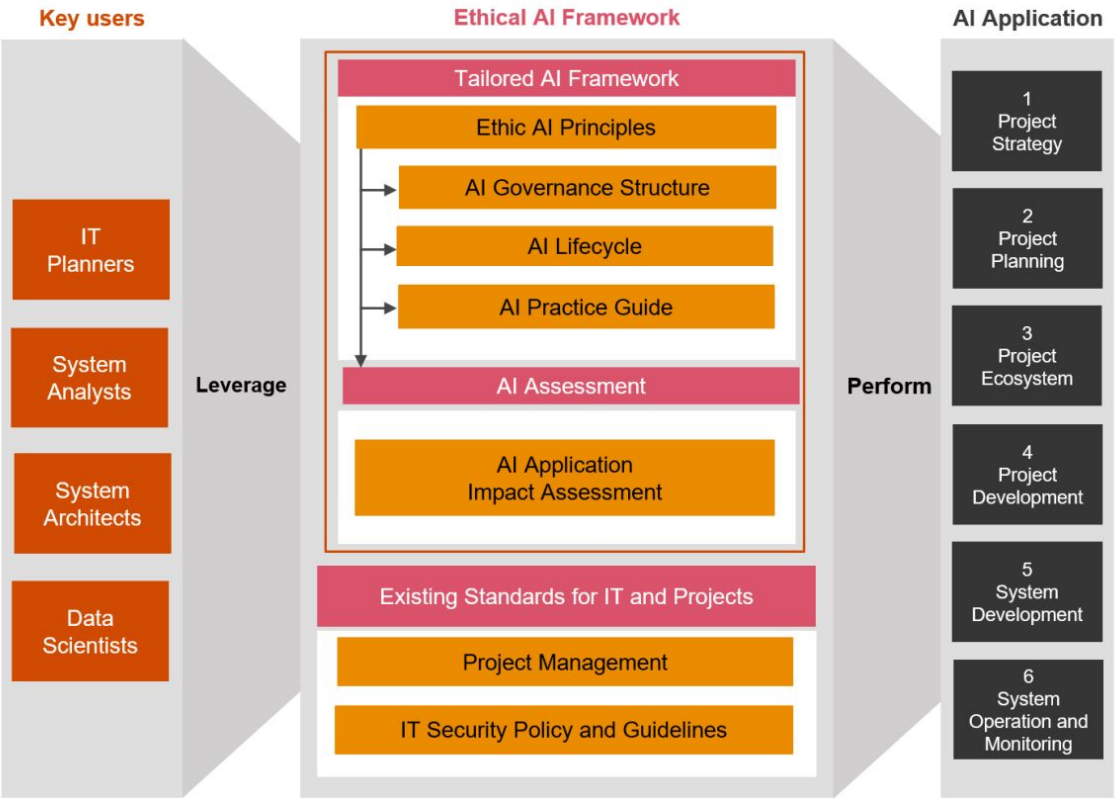
- AI enabled video games
- Spam filters

Any monitor on the use/development of AI tools in Hong Kong?

Ethical Artificial Intelligence Framework

The Digital Policy Office (DPO) issued the framework on July 2024.

It is the internal adoption within the Government regarding the applications of AI and big data analytics.



Reference:
https://www.digitalpolicy.gov.hk/en/our_work/data_governance/policies_standards/ethical_ai_framework/

Ethical AI Principles

1. **Transparency and Interpretability:** Organisations should be able to explain decision-making processes of the AI applications to humans in a clear and comprehensible manner.
2. **Reliability, Robustness and Security:** Like other IT applications, AI applications should be developed such that they will operate reliably over long periods of time using the right models and datasets while ensuring they are both robust (i.e. providing consistent results and capable to handle errors) and remain secure against cyber-attacks as required by the relevant legal and industry frameworks.
3. **Fairness:** The recommendation/result from the AI applications should treat individuals within similar groups in a fair manner, without favouritism or discrimination and without causing or resulting in harm. This entails maintaining respect for the individuals behind the data and refraining from using datasets that contain discriminatory biases.

Ethical AI Principles

4. **Diversity and Inclusion:** Inclusion and diverse usership through the AI application should be promoted by understanding and respect the interests of all stakeholders impacted.
5. **Human Oversight:** The degree of human intervention required as part of AI application's decision-making or operations should be dictated by the level of the perceived severity of ethical issues.
6. **Lawfulness and Compliance:** Organisations responsible for an AI application should always act in accordance with the law and regulations and relevant regulatory regimes.
7. **Safety:** Throughout their operational lifetime, AI applications should not compromise the physical safety or mental integrity of mankind.

Ethical AI Principles

8. **Data Privacy:** Individuals should have the right to be informed of the purpose of the collection of the personal data, be assured that the data is not kept longer than its necessary, be assured that the data take all practicable steps to protect the personal data.
9. **Accountability:** Organisations are responsible for the moral implications of their use and misuse of AI applications. There should also be a clearly identifiable accountable party, be it an individual or an organisational entity (e.g. the AI solution provider).
10. **Beneficial AI:** The development of AI should promote the common good.
11. **Cooperation and Openness:** A culture of multi-stakeholder open cooperation in the AI ecosystem should be fostered.
12. **Sustainability and Just Transition:** The AI development should ensure that mitigation strategies are in place to manage any potential societal and environmental system impacts.

AI Governance - Lines of Defence Model

First Line
Project Team

AI Application Development

- AI application developers or owners must follow appropriate standards
- AI application aligned with overall strategy

Process Documentation

- Project teams document the AI assessment throughout the AI lifecycle

Risk Assessment

- Evaluate potential risk associated with AI application through AI Assessment

Second Line
Project Steering Committee (PSC)/ Project Assurance Team (PAT)

Standards & Specifications

- Organisation leaderships define the acceptance criteria for AI application and overall requirement

Independent Review & Challenge

- Independent review group evaluates AI application's performance
- Reviews of controls and assessments
- Ongoing monitoring
- Comment/input for high risk cases

Final Approval

- PSC/PAT signoff prior to AI application delivery

Third Line
IT board/
Chief Information Officer (CIO)

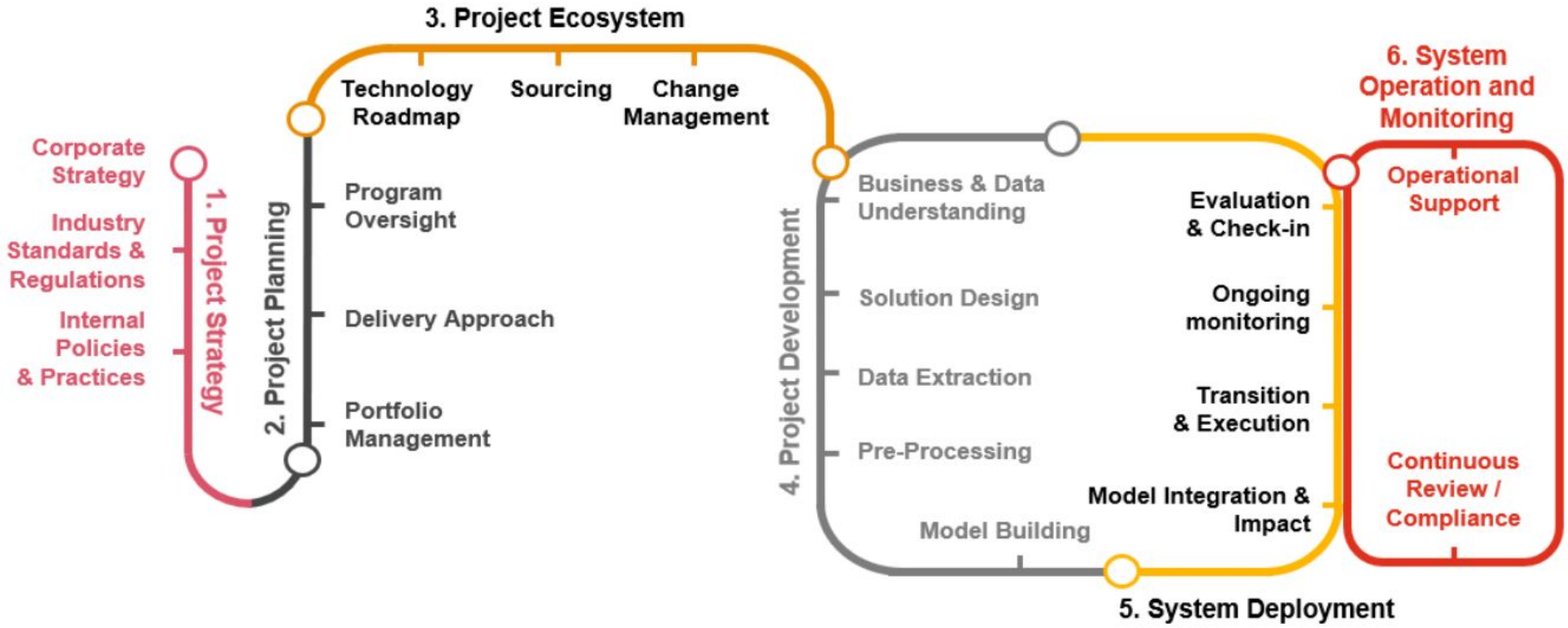
For high-risk AI application:

- Senior management directive and review
- Ongoing monitoring

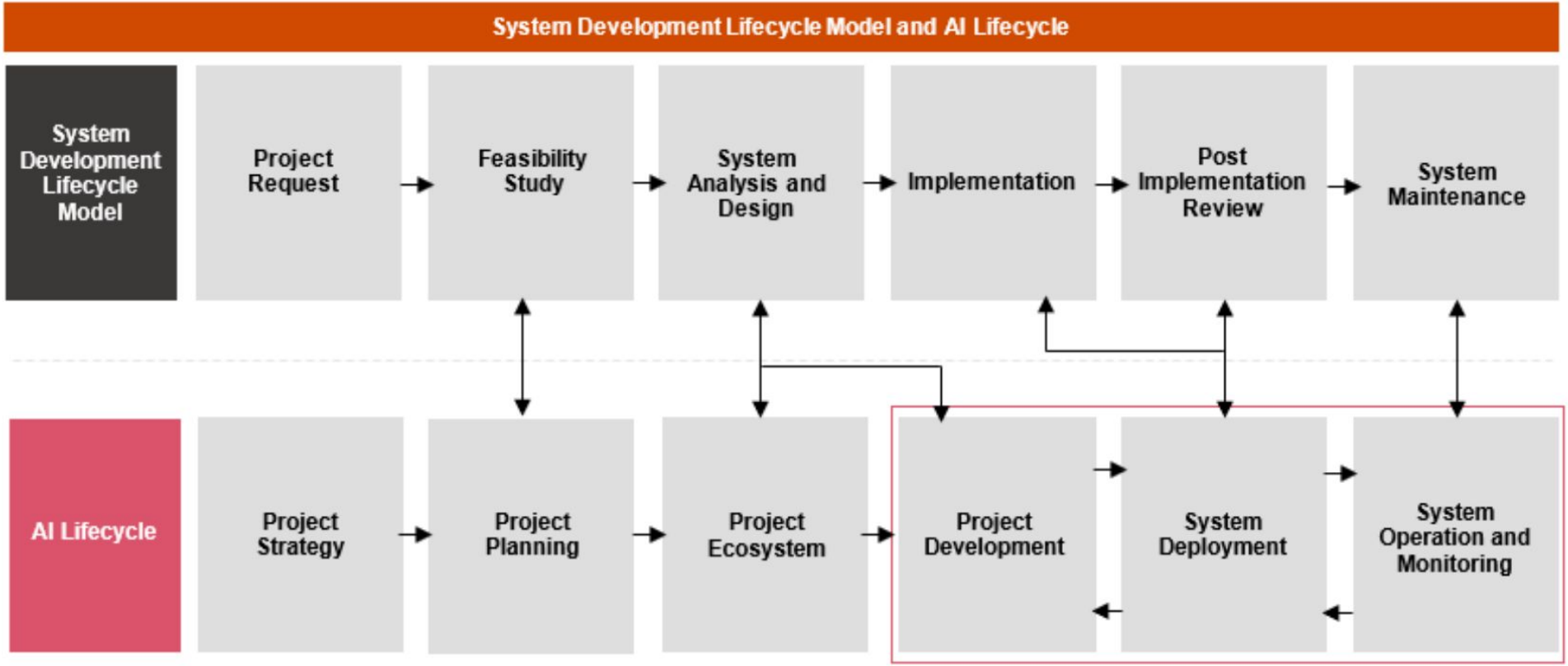
Ethical AI Committee (optional)

- Act as advisors to the IT board

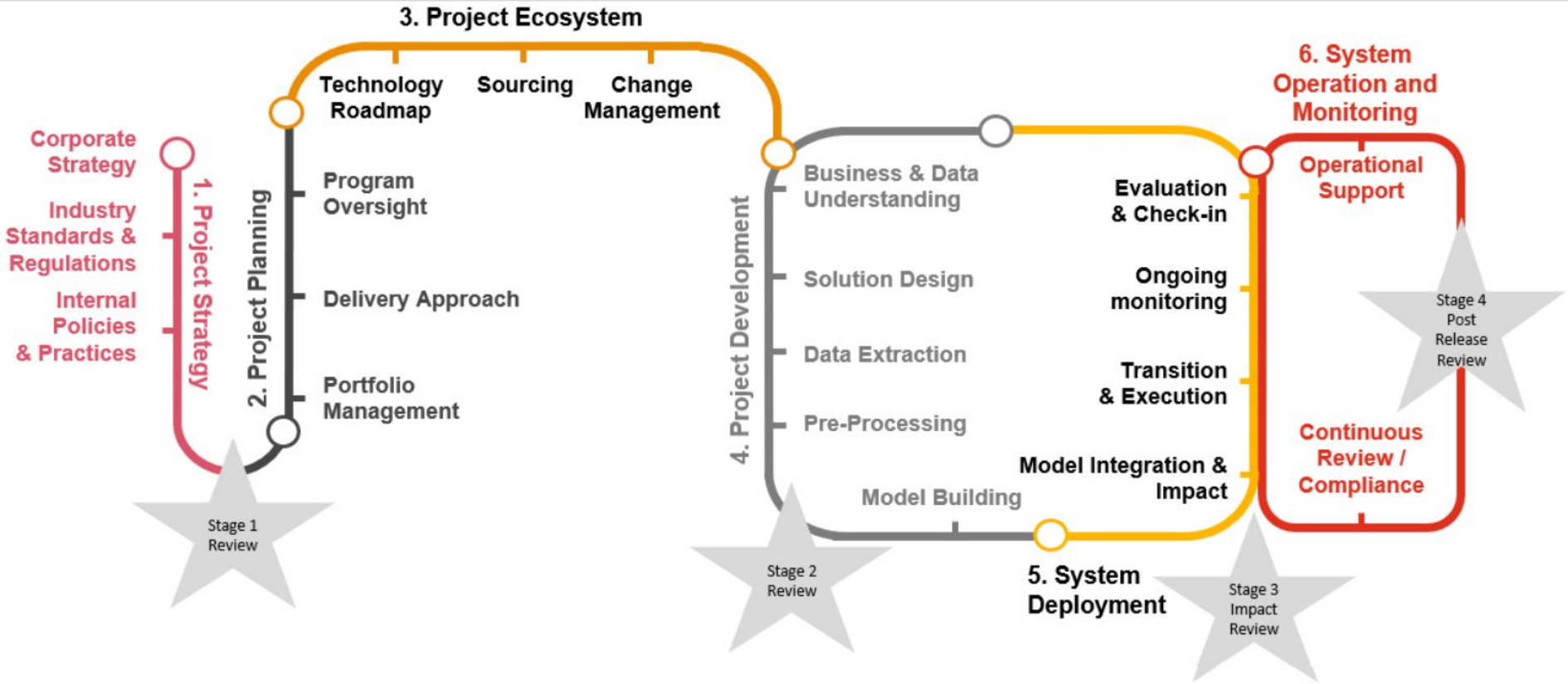
Overview of the AI Lifecycle



AI Lifecycle aligned with a System Development Lifecycle Model



Stages for AI Application Impact Assessment



Example: Stages for AI Application Impact Assessment

1. During the project request, the project team categorises the project as high risk or not based on the “Risk Gating questions”.
2. During the system analysis and design and the in-house implementation, the project team conduct AI application impact assessment.
3. During the system maintenance, the maintenance team reviews and updates the AI application impact assessment, and escalate any significant issues to the maintenance board or IT board.
4. … etc

Remarks: The definition of Risk Tier is different from EU Act

Risk Tier	Definition	Regulatory Strategy
Unacceptable Risk	Systems posing existential threats (e.g., uses causing harm or affecting human safety, subliminal manipulation)	• Full prohibition • Legal liability for development/deployment
High Risk	Critical infrastructure systems (e.g., healthcare diagnostics, autonomous vehicles)	• Conformity assessments • Human-in-the-loop requirements • Real-time monitoring
Limited Risk	Systems with moderate societal impact (e.g., recruitment tools, educational AI)	• Transparency obligations • User opt-out mechanisms • Annual compliance audits
Low Risk	Minimal-risk applications (e.g., spam filters, creative tools)	• Self-certification

Any monitor on the use/development of AI tools in the Mainland?

The Governance Requirements for GenAI in the Mainland

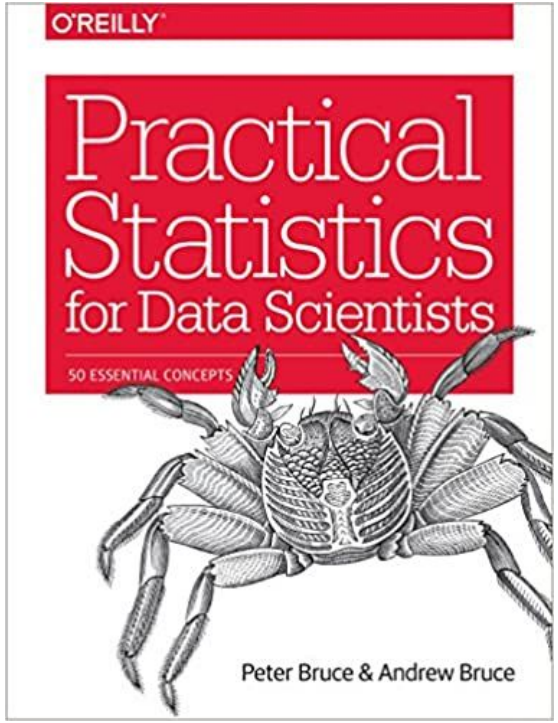
- The Mainland issued the “Global AI Governance Initiative”.
- The “Interim Measures for the Management of Generative AI Services” is promulgated under the legal framework for the Cybersecurity Law, Data Security Law and Personal Information Protection Law.
- The Measures were effect in August 2023.

Reference Books

Practical Statistics for Data Scientists: 50 Essential Concepts

Chapters:

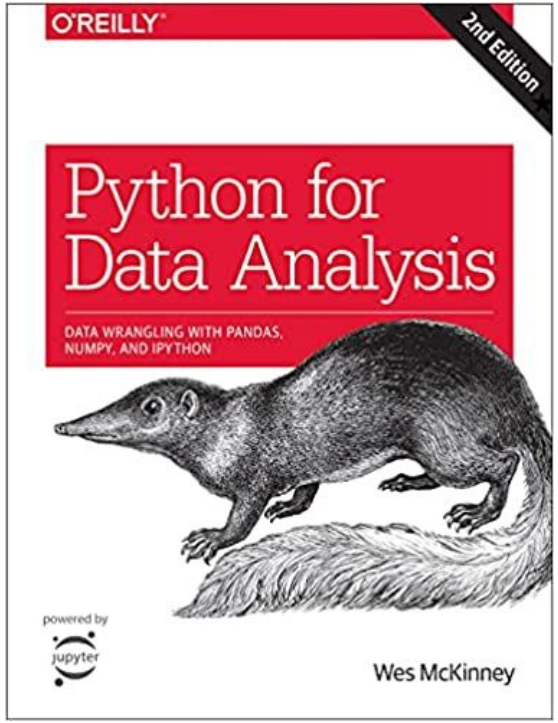
1. Exploratory Data Analysis
2. Data and Sampling Distributions
3. Statistical Experiments and Significance Testing
4. Regression and Prediction
5. Classification
6. Statistical Machine Learning
7. Unsupervised Learning



Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython

Chapters

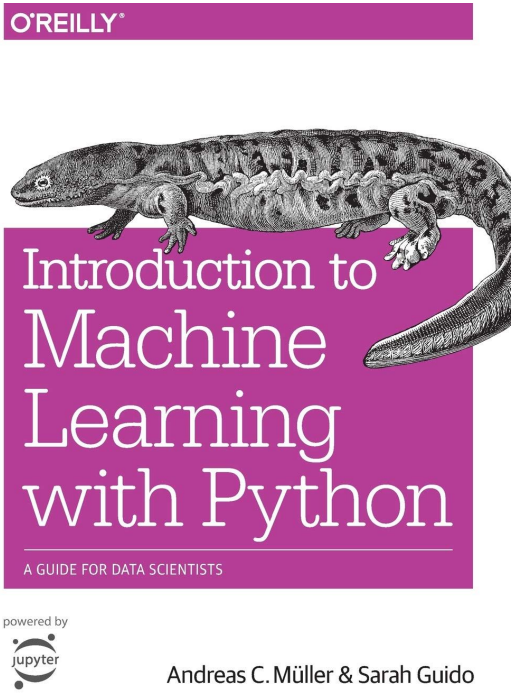
1. Preliminaries
2. Python language basics, IPython, and Jupyter notebooks
3. Built-in data structures, functions, and files
4. **NumPy basics: arrays and vectorized computation**
5. **Getting started with pandas**
6. **Data loading, storage, and file formats**
7. **Data cleaning and preparation**
8. **Data wrangling: join, combine, and reshape**
9. **Plotting and visualization**
10. **Data aggregation and group operations**
11. Time series
12. Advanced pandas
13. Introduction to modeling libraries in Python
14. Data analysis examples
15. Advanced NumPy
16. More on the IPython system



Introduction to Machine Learning with Python: A Guide for Data Scientists

Chapters:

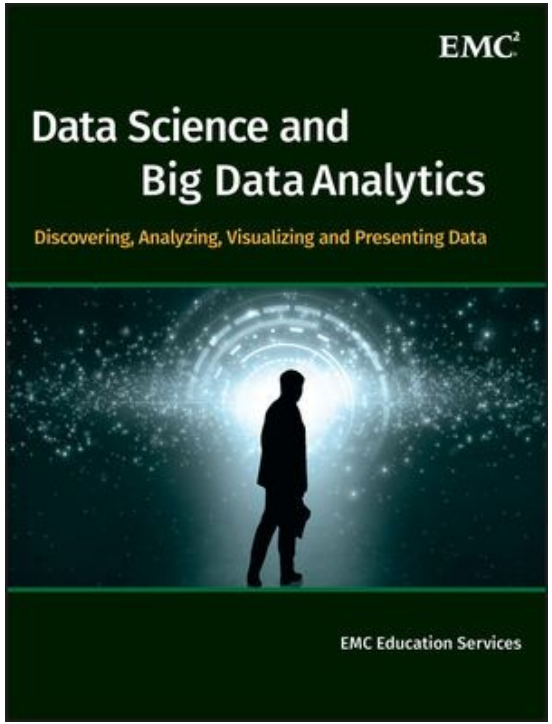
1. Introduction
2. Supervised Learning
3. Unsupervised Learning and Preprocessing
4. Representing Data and Engineering Features
5. Model Evaluation and Improvement
6. Algorithm Chains and Pipelines
7. Working with Text Data
8. Wrapping Up



Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data

Chapters:

1. Introduction to Big Data Analytics
2. Data Analytics Lifecycle
3. Review of Basic Data Analytic Methods Using R
4. Advanced Analytical Theory and Methods: Clustering
5. Advanced Analytical Theory and Methods: Association Rules
6. Advanced Analytical Theory and Methods: Regression
7. Advanced Analytical Theory and Methods: Classification
8. Advanced Analytical Theory and Methods: Time Series Analysis
9. Advanced Analytical Theory and Methods: Text Analysis
10. Advanced Analytics - Technology and Tools: MapReduce and Hadoop
11. Advanced Analytics - Technology and Tools: In-Database Analytics
12. The Endgame, or Putting It All Together



Deep Learning

Chapters:

1. Introduction

Part I: Applied Math and Machine Learning Basics

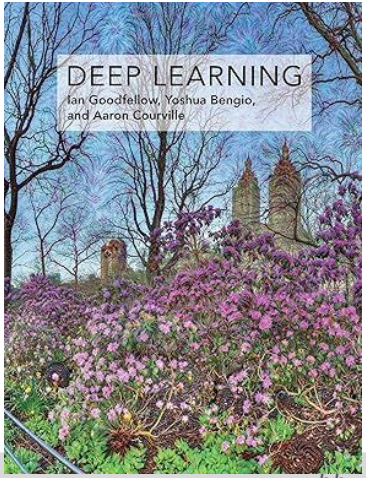
2. Linear Algebra
3. Probability and Information Theory
4. Numerical Computation
5. Machine Learning Basics

Part II: Modern Practical Deep Networks

6. Deep Feedforward Networks
7. Regularization for Deep Learning
8. Optimization for Training Deep Models
9. Convolutional Networks
10. Sequence Modeling: Recurrent and Recursive Nets
11. Practical Methodology
12. Applications

Part III: Deep Learning Research

13. Linear Factor Models
14. Autoencoders
15. Representation Learning
16. Structured Probabilistic Models for Deep Learning
17. Monte Carlo Methods
18. Confronting the Partition Function
19. Approximate Inference
20. Deep Generative Models



Reference: https://www.deeplearningbook.org/

Python Programming Environment

What is Python?

- An high-level interpreted programming language
- Created by Guido van Rossum in 1991
- Emphasizes code readability and flexibility (See Python's Design Philosophy)
- Current stable versions: **Python 3.13 (Version 3)**
  - *All assignments are graded under Anaconda Python 3.13*

First Python Program: Hello World!

- Hello World in Python

```
C:\Users\yipki>python
Python 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37:03) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello World!")
Hello World!
>>> |
```

- Type python (or python3) to invoke the Python interpreter
- `print()` will output the arguments to standard output

Python Programs

- Python programs (or scripts) are commonly named using the .py extension, and are called **modules**
- A "hello world" script helloworld.py:

```
helloworld.py
1  print("Hello World")
```

- Executing the script:

```
C:\Users\ksyip>python helloworld.py
Hello World

C:\Users\ksyip>_
```

What do people use Python for?

Python is a general purpose programming language and are widely used in different domains. (See Python Success Stories)

- Web and Internet applications backend (e.g. Youtube, Dropbox, Reddit)
- Scientific computing
- Data science and machine learning (e.g. PyTorch)
- Data visualization
- Financial Analysis
- … etc

Installing Python

- Available on Linux / Mac / Windows (https://www.python.org/downloads/)

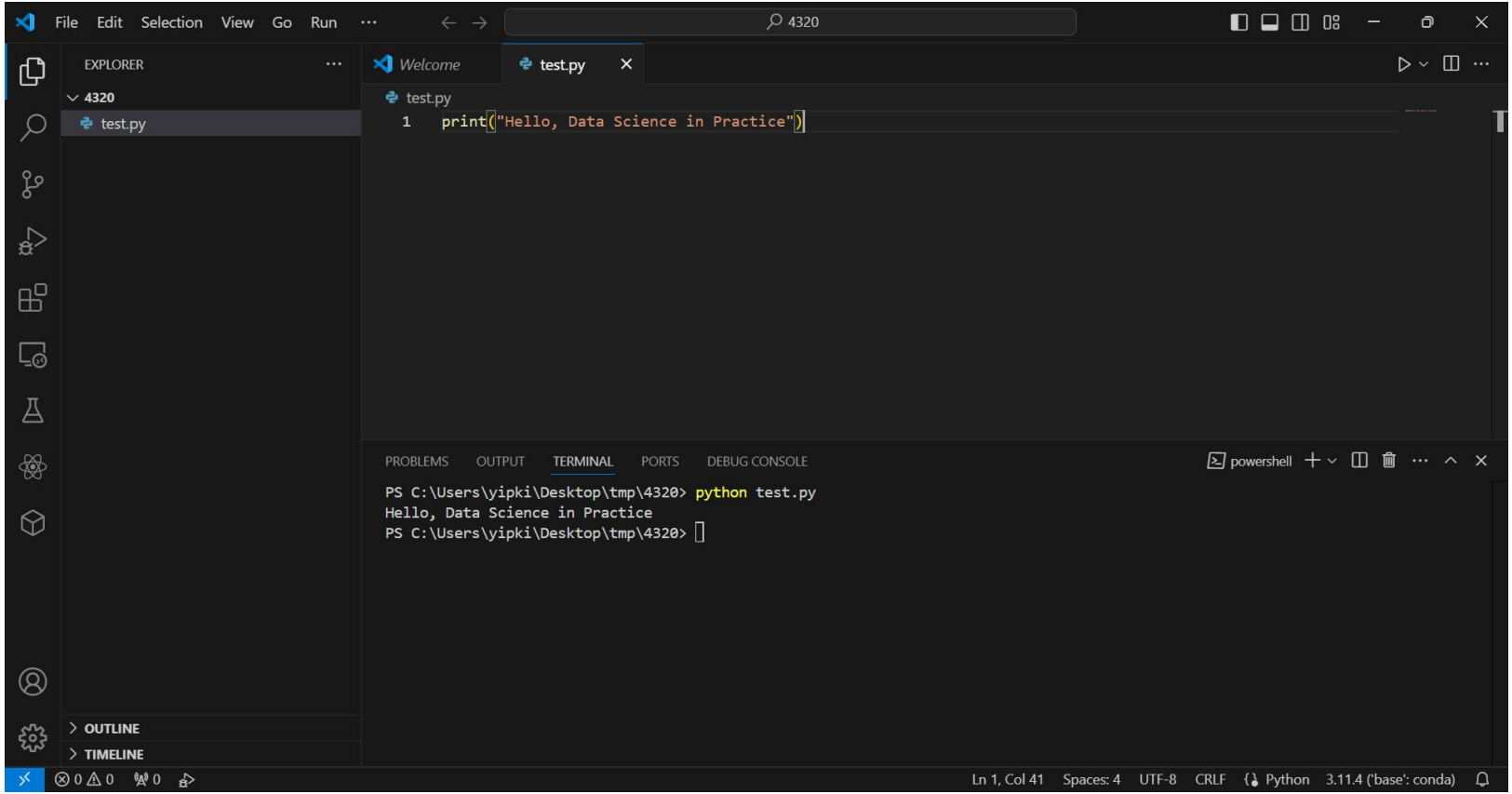
- However, I recommend to use the "Anaconda Python 3.13" for this course

  Advantages of Anaconda:

1. Free, open-source
2. Over 1500 Python data science packages
3. Simplifying the package management and deployment
4. Ongoing development, maintenance and upgrade

- IDEs for Python programming:
  - MS Visual Studio Code (Free and open source)

[Recommended] Using Python in VSCode



Python Documentations and Tutorials

Documentations

1. https://en.wikipedia.org/wiki/Python_(programming_language)
2. https://docs.python.org/3/

Tutorials

1. https://www.learnpython.org/en/Welcome
2. https://www.w3resource.com/python-exercises/
3. https://docs.python.org/3/tutorial/
4. https://www.w3schools.com/python/default.asp